👤 **Enron Power Trading Department**

Team ID: enron_power_trading_department

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Nitor Energy's Annual Case Competition - Instructions

Competition Overview

Welcome to **Nitor Energy's Annual Case Competition**! Your task is to predict the intraday price for electricity market conditions across different markets and time periods.

Objective

Build a model that accurately predicts the **intraday price** for electricity market conditions across different markets and time periods.

What is the intraday price?

In electricity markets, power is bought and sold in two key stages:

1. **Day-ahead market**: The day before delivery, producers and consumers trade power for each hour of the next day. This sets the **day-ahead price** — think of it as the "planned" price.
2. **Intraday / real-time market**: On the actual delivery day, things change — wind blows more or less than expected, a

power plant trips, demand spikes. So on the day of delivery, the price outlook may have changed..

Your task is to predict the intraday prices for the hourly delivery of power. This competition simulates how energy traders at Nitor Energy operate on a daily basis.

Evaluation Metric

Submissions are scored using **RMSE (Root Mean Squared Error)** computed on a **public subset** of the test data. Lower scores are better.

RMSE Formula:

```
RMSE = sqrt(mean((y_true - y_pred)^2))
```

⚠️ **Important**: The public leaderboard is based on only a portion of the test data. Your final ranking will be determined by performance on the full test set after the competition ends.

Submission Format

Your submission **must** be a CSV file with exactly 2 columns and 13,098 rows.

Required Format

Column	Description	Type
`id`	Row identifier (133627-146778)	Integer
`target`	Your prediction	Float

Example Submission (first 5 rows)

```
id,target
133627,24.206
133628,4.691
133629,-19.376
133630,24.162
133631,20.963
```

Validation Requirements

✅ **Columns**: Must include exactly `id` and `target` (case-sensitive)
✅ **Row Count**: Must have exactly 13,098 rows
✅ **id**: Must match the IDs from the sample submission file exactly
✅ **target**: All predictions must be numeric (no NaN or infinite values)
✅ **File Format**: CSV with comma delimiter and header row
✅ **No Duplicates**: Each ID must appear exactly once

Creating a Submission (Python)

Step 1: Load Sample Submission

```
import pandas as pd
import numpy as np

# Load the sample submission to get correct forma
sample_submission = pd.read_csv("sample_submissio
print(f"Submission shape: {sample_submission.shap
print(sample_submission.head())
# Output: (13098, 2) - 13,098 rows, 2 columns
```

Step 2: Load Test Data

```
# Load test data (without targets)
test_data = pd.read_csv("test_for_participants.cs
print(f"Test data shape: {test_data.shape}")
print(test_data.columns)
# Note: test_data includes all features but NOT t
```

Step 3: Generate Predictions

```
# Load your trained model
# model = load_model("my_trained_model.pkl")

# Generate predictions
predictions = sample_submission.copy()

# Example: Replace with your model's predictions
predictions['target'] = 0.0  # Placeholder - use 
# Or use your trained model:
# predictions['target'] = model.predict(test_data
```

Step 4: Validate Your Submission

```
# Check format
assert list(predictions.columns) == ['id', 'targe
assert len(predictions) == 13098, f"Wrong row cou
assert predictions['id'].min() == 133627, "IDs mu
assert predictions['id'].max() == 146778, "IDs mu
assert predictions['target'].notna().all(), "No N
assert np.isfinite(predictions['target']).all(),

print("✅ Validation passed!")
```

Step 5: Save Submission

```
# Save to CSV
predictions.to_csv("my_submission.csv", index=Fal
print(f"Submission saved: {predictions.shape}")
```

Competition Rules

Submission Limits

- **Rate Limit**: 1 submission per hour per team
- **Team Collaboration**: Each team has one shared login
- **Concurrent Logins**: Only one person per team can be logged in at a time

Scoring System

- Each submission is scored against the public test subset
- Only your **best score** appears on the leaderboard
- Each new submission updates your submission count and timestamp
- Your leaderboard position is determined by your best public RMSE score
- **The number of submissions you make has no consequence** — submit as often as the rate limit allows

Final Submission ⚠️ Very Important

For your final submission to be deemed **valid**, you must send an email to mkp@nitorenergy.com before the competition deadline (**February 22, 2026 at 22:00 CET**) with the following **3 things**:

1. **Team name**
2. **Submission CSV** attached to the email
3. **A GitHub link** to the repository containing the code that produced the above submission
    - We don't judge you based on your code, but we need to verify that no cheating occurred.

❗ **Without this email, your submission will not count towards the final ranking.**

Tips for Success

Validation Strategy

⚠️ **Avoid Overfitting to Public Leaderboard**

- Use robust local validation (time-based CV)
- Don't tune excessively to public scores
- Ensure model generalizes to unseen data

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For technical support or questions, contact: mkp@nitorenergy.com